

Day 05

Cloud Tools & Cloud Applications

Cloud Computing and Big Data Analytics

Graduate Studies — Digital Science

Today's Agenda

Lesson	Topic	Focus
Lesson 1	Cloud Tools	Build, manage, monitor, secure cloud environments
Lesson 2	Cloud Applications	Software products running on cloud infrastructure

Mental Model

Cloud Tools = the vehicles (instruments)

Cloud Applications = the destinations (value delivered)

LESSON 1: Cloud Tools

The instruments and services that power cloud environments

What Is a Cloud Tool?

Definition

Any utility, platform feature, or managed service that helps engineers provision, configure, deploy, monitor, or secure cloud resources.

Kitchen Analogy

Cloud apps = dishes served to customers

Cloud tools = ovens, knives, thermometers in the kitchen

Six Categories of Cloud Tools

1

Infrastructure Provisioning

Create VMs, networks, storage

2

Orchestration & Automation

Coordinate workflows, auto-scale

3

Monitoring & Observability

Track performance, alerts, logs

4

Security & Identity

Access control, encryption

5

Data Engineering

Move, transform, store data

6

Cost Management

Track spending, optimise

Infrastructure Provisioning — The Big Three

Provider	Console	CLI Tool
AWS (Amazon Web Services)	Management Console	aws CLI
Microsoft Azure	Azure Portal	az CLI
GCP (Google Cloud Platform)	Cloud Console	gcloud CLI

Infrastructure as Code (IaC)

Describe infrastructure in text files — version-controlled like app code.

Tool	Scope	Language
Terraform (HashiCorp)	Multi-cloud	HCL (HashiCorp Configuration Language)
AWS CloudFormation	AWS only	JSON / YAML templates

- Repeatability — identical environments every time
- Auditability — all changes tracked in version control
- Speed — provision 10 VMs in one command

KEY TAKEAWAY

Infrastructure provisioning tools let you create cloud resources programmatically. IaC (Infrastructure as Code) makes this process repeatable, auditable, and fast. For data scientists, this means spinning up training environments in minutes, not days.

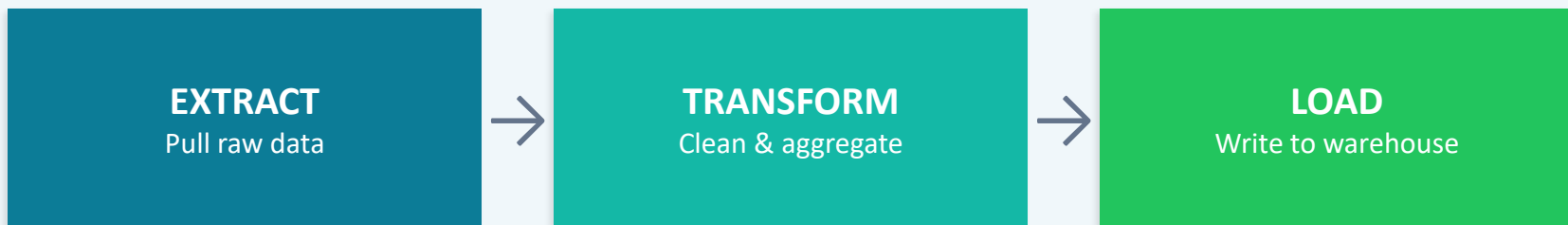
Orchestration & Automation

Tool	Purpose	Maintained By
Kubernetes (K8s)	Container orchestration	CNCF
Docker	Container runtime	Docker, Inc.
Apache Airflow	DAG-based workflow scheduling	Apache Software Foundation
Ansible	Configuration management	Red Hat

Apache Airflow & the ETL Pattern

Why Airflow matters for Data Science

- Define a chain of tasks as a DAG (Directed Acyclic Graph)
- Schedule the entire chain to run automatically (e.g., nightly)
- Example: Extract → Clean → Train model → Push predictions



KEY TAKEAWAY

Orchestration tools like Kubernetes and Apache Airflow coordinate multi-step workflows automatically. For data scientists, Airflow is essential — it turns manual nightly scripts into reliable, self-running ETL pipelines.

Monitoring — Three Pillars of Observability

METRICS

CPU, memory, latency
(numbers)

LOGS

Event records
(text)

TRACES

Request journeys
(paths)

Running cloud workloads without monitoring = driving at night with the headlights off.

Monitoring Tools Landscape

Tool	Type
Prometheus	Open-source metrics + alerting
Grafana	Open-source dashboarding + visualisation
AWS CloudWatch	AWS-native monitoring
Datadog	Commercial full-stack observability
ELK Stack	Log management (Elasticsearch + Logstash + Kibana)

KEY TAKEAWAY

Monitoring tools collect metrics, logs, and traces to keep cloud systems healthy. When your ML API slows down at 3 AM, alerts from Prometheus + Grafana wake the on-call engineer before customers notice.

Cloud Security & Identity

Shared Responsibility Model

PROVIDER secures infrastructure
(physical data centres, hypervisors)

**YOU secure data, identities,
and application logic**

Concept	What It Does
IAM (Identity & Access Management)	Controls who can do what on which resource
MFA (Multi-Factor Authentication)	Second proof of identity beyond password
VPC (Virtual Private Cloud)	Isolated network segment in public cloud
KMS (Key Management Service)	Manages encryption keys

Principle of Least Privilege

Grant the minimum permissions necessary. Nothing more.

Role	Allowed Actions	Decision
Student (15 users)	READ only	GET ✓ PUT ✗ DELETE ✗
Professor (1 user)	Full access	GET ✓ PUT ✓ DELETE ✓
Guest	No policy	ALL DENIED

KEY TAKEAWAY

Cloud security follows the shared responsibility model. Configure IAM policies correctly and apply the principle of least privilege — every request not explicitly allowed is denied by default.

Cloud Cost Management

- Cloud charges by usage — CPU-hours, GB stored, network egress
- A forgotten GPU instance can cost thousands in a weekend
- Always estimate BEFORE you deploy

Tool	Provider
AWS Cost Explorer	AWS
Azure Cost Management	Azure
GCP Billing Reports	GCP
Infracost	Multi-cloud (open-source)

Cost Estimate — NLP Training Project

Resource	Quantity	Unit Price	Subtotal
GPU Compute (p3.2xlarge)	120 hrs	\$3.06/hr	\$367.20
SSD Storage	500 GB	\$0.10/GB/mo	\$50.00
Network Egress	50 GB	\$0.09/GB	\$4.50
TOTAL			\$421.70



Savings tip: Spot / preemptible instances can reduce GPU costs by 60–90%

KEY TAKEAWAY

Always estimate before you deploy. Cloud cost management tools prevent budget surprises. The biggest risk for data scientists is leaving GPU instances running after a training job finishes.

Lesson 1 Recap — Cloud Tools

#	Category	Key Idea
1	Infrastructure Provisioning	Automate with IaC
2	Orchestration & Automation	Coordinate with DAGs
3	Monitoring & Observability	Metrics + Logs + Traces
4	Security & Identity	IAM + least privilege
5	Data Engineering	ETL pipelines at scale
6	Cost Management	Estimate, track, optimise

LESSON 2: Cloud Applications

Software products running on cloud infrastructure, accessed via the internet

What Is a Cloud Application?

A software system designed for the internet, running on cloud infrastructure, maintained by the provider — the user installs nothing locally.

Synonym: SaaS (Software as a Service)

Examples you already use: Google Workspace, Microsoft 365, Slack, Zoom, Notion

Cloud Applications vs. Traditional Software

Feature	Traditional	Cloud Application
Installation	Per-machine	None (browser)
Updates	Manual download	Automatic
Collaboration	Email files	Real-time co-editing
Data Storage	Local hard drive	Provider's cloud
Scaling	Limited to hardware	Elastic on demand
Pricing	One-time licence	Subscription

KEY TAKEAWAY

Cloud applications eliminate installation, enable real-time collaboration, and scale on demand. For data-science teams this means faster iteration and no 'which version is the latest?' confusion.

Cloud Storage & Databases

Object Storage

- Amazon S3 (Simple Storage Service)
- Google Cloud Storage
- Azure Blob Storage

Data Warehouses

- Snowflake
- Amazon Redshift
- Google BigQuery
- MongoDB Atlas (NoSQL)

The Data Lake Architecture

Zone	Contents	Access
RAW	Immutable originals, date-stamped	Ingest scripts only
PROCESSED	Cleaned & transformed datasets	Data engineers
CURATED	Analysis-ready tables & dashboards	All researchers (read-only)

KEY TAKEAWAY

Cloud storage handles petabytes without managing physical drives. The data lake pattern (Raw → Processed → Curated) is the standard architecture for organising large-scale research data.

Big Data Processing

Core Idea: Horizontal Scaling

Instead of buying a bigger computer → add more computers and split the job.

Platform	Processing Model
Apache Spark	In-memory distributed computing
Apache Hadoop	Disk-based MapReduce
Google Dataflow	Serverless stream + batch (Apache Beam)
Amazon EMR	Managed Hadoop/Spark clusters

The MapReduce Pattern



Classic example: Word Count — MAP emits (word, 1), REDUCE sums all 1s per word

KEY TAKEAWAY

When data exceeds one machine's capacity, distributed frameworks like Spark split the workload across clusters. The MapReduce pattern (Map → Shuffle → Reduce) is the foundational algorithm behind most big data tools.

Cloud ML Platforms

Platform	Provider
Amazon SageMaker	AWS
Google Vertex AI	GCP
Azure Machine Learning	Azure
Databricks	Multi-cloud
Hugging Face	Open-source hub

What they provide: Managed GPU notebooks, experiment tracking, model versioning, one-click deployment as REST APIs

The Cloud ML Workflow



Client sends "My order is late!" → API returns: NEGATIVE

KEY TAKEAWAY

Cloud ML platforms handle the infrastructure complexity of training and deploying models. You focus on the data science; the platform handles GPUs, scaling, and serving.

Collaboration & Productivity Apps

Application	Category
Google Workspace	Docs, Sheets, Slides, Drive, Meet
Microsoft 365	Word, Excel, Teams, OneDrive
Slack	Team messaging + integrations
Notion	All-in-one workspace
Trello	Kanban-style project boards
Miro	Collaborative whiteboard

KEY TAKEAWAY

Cloud collaboration apps power the daily workflow of every data-science team: shared drives for datasets, messaging for standups, task boards for milestones, and shared decks for presentations.

Serverless Computing

Provider dynamically allocates resources. You pay only for actual compute time.

Service	Provider
AWS Lambda	AWS
Azure Functions	Azure
Google Cloud Functions	GCP
Cloudflare Workers	Cloudflare

Serverless — Event-Driven Model

1

EVENT
occurs

File upload, HTTP req, timer

2

FUNCTION
triggered

Auto-invoked by provider

3

CODE
executes

Process + return result

4

RESOURCES
released

\$0.00 when idle

Ideal for: image processing, webhooks, scheduled tasks, lightweight APIs

KEY TAKEAWAY

Serverless computing is the ultimate pay-per-use model. Write a function, define a trigger, and let the cloud handle everything else. Perfect for event-driven data processing tasks.

Lesson 2 Recap — Cloud Applications

#	Domain	Key Examples
1	Storage & Databases	S3, BigQuery, Snowflake
2	Big Data Processing	Spark, Hadoop, Dataflow
3	ML / AI Platforms	SageMaker, Vertex AI, Databricks
4	Collaboration	Google Workspace, Slack, Notion
5	Serverless Computing	Lambda, Cloud Functions

Full Lecture Summary

Cloud Tools (Lesson 1)

The operational machinery — provisioning, orchestration, monitoring, security, data engineering, cost management.

Cloud Applications (Lesson 2)

The end-user products — from data warehouses to ML platforms to collaboration suites.

Cloud tools BUILD & MAINTAIN the environment. Cloud applications RUN ON TOP of it.

Key Abbreviations

Abbrev.	Full Form	Abbrev.	Full Form
IaC	Infrastructure as Code	SaaS	Software as a Service
IAM	Identity & Access Mgmt	API	Application Programming Interface
ETL	Extract, Transform, Load	VM	Virtual Machine
DAG	Directed Acyclic Graph	ML	Machine Learning
VPC	Virtual Private Cloud	GPU	Graphics Processing Unit

Homework Assignment

1. Pick one cloud tool and one cloud application from today's lecture
2. Sign up for a free tier (most providers offer generous free tiers)
3. Complete a short hands-on exercise
4. Document what you learned in a shared cloud document

Questions?

Day 05 — Cloud Tools & Cloud Applications

Cloud Computing and Big Data Analytics

Graduate Studies in Digital Science

Material by Cyber.SoHo — All Rights Reserved